Contact

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Education

Oct 2024 – Oct 2026
M.Sc. Elektromobilität-ACES
Friedrich-Alexander-Universität Erlangen-Nürnberg
GPA: 1.9

Jul 2018 – Sep 2022
B.Tech in Mechanical Engineering
SVNIT Surat
GPA: 2.4

Technical Skills

ROS2 & Robotics
ROS2 (Jazzy), MoveIt2, ros2_control, TF2, Nav2, Isaac Sim, ROS2 launch files & nodes

Programming
Python, C++, PyTorch, OpenCV

DevOps & Deployment
Docker, Git, CI/CD pipelines, real-time inference

Methodologies
Agile, Life-cycle analysis, Risk assessment

CAD & Simulation
SolidWorks, CATIA V5, Creo, Ansys Fluent, OpenFOAM, 3D Printing (FDM)

Languages

English – C1 (Fluent)

German – B1 (Intermediate)

ARAVIND PALAKKAL

Professional Summary

M.Sc. Elektromobilität-ACES student at FAU Erlangen-Nürnberg with hands-on experience building ROS2 pipelines, 7-DOF humanoid arms, and VLA models at a robotics startup. Skilled in sim-to-real validation using Isaac Sim, MoveIt2, and Docker-based deployment.

Professional Experience

JARM.AI GmbH *Nov 2025 – Apr 2026*
Werkstudent – Robotics Engineer

- Developed full **ROS2 (Jazzy)** software stack for 7-DOF humanoid arm across **5** packages: custom URDF models, **`ros2_control` hardware interfaces, MoveIt2 motion planning pipeline, and Isaac Sim integration for end-to-end sim-to-real validation**.
- Refactored ROS2 driver abstraction layer, reducing system bring-up time by **30%**; documented implementation in internal technical guides distributed to the engineering team.
- Containerized robotic software stack using **Docker**, cutting environment setup time by **50%** and enabling consistent **CI/CD** deployment across all development environments.
- Designed a 7-DOF humanoid arm and robotic hand in **SolidWorks/Onshape** across **6+** iterations; produced 3D-printed (FDM) prototype components for fit and assembly validation.
- Trained and optimized **Vision-Language-Action (VLA) models** on the S101 robotic arm; iterated on datasets and evaluation pipeline, improving task success rate from **<50% to >80%**.

Arise Ports & Logistics *Sep 2022 – Nov 2023*
Graduate Engineering Trainee (Railway Operations)

- Built **Arduino-based** vibration analysis system for SD40 diesel locomotives (EMD645); predictive maintenance model reduced unplanned downtime by **47%**.
- Led data collection and analysis across **120+** maintenance reports, improving decision turnaround time by **15%**.
- Applied SMDE principles to cut locomotive maintenance cycle time by **2 hours**, improving equipment availability by **10%**.
- Managed **25+** overhaul schedules and **12+** risk assessments, supporting data-driven maintenance planning.

Projects

- **Autonomous Warehouse AMR — ROS2 + Nav2 + Docker**
  `github/ros2_logistics_amr`
  Built a full autonomous mobile robot stack for warehouse logistics: custom URDF/XACRO model, ROS2 Jazzy + Gazebo Harmonic simulation, SLAM Toolbox mapping, Nav2 autonomous navigation with AMCL localization, and `ros2_control` diff-drive interface. Containerised with Docker; CI/CD pipeline via GitHub Actions (**passing**). Robot navigates **4 waypoints** autonomously without manual intervention.
- **Haptic Interface for UR Robot:** Developed force-feedback haptic interface for a 6-DOF Universal Robot integrated with ROS for real-time control of delicate object manipulation.
- **ML Model for Auxetic Structure Optimization:** Designed and simulated **500+** Auxetic structures in SolidWorks/ANSYS Fluent; trained regression model achieving **96.2%** accuracy in deformation prediction.
- **RNN Airfoil Optimization (OpenFOAM):** Currently running CFD simulation of dual-airfoil systems and training a RNN model to predict stall angles and optimize aerodynamic configurations.
- **Wedge-Based Combat Robot (Valyrion Robotics Club):** Designed chassis on Creo through 6 iterative feedback rounds; SolidWorks prototypes for manufacturing validation.